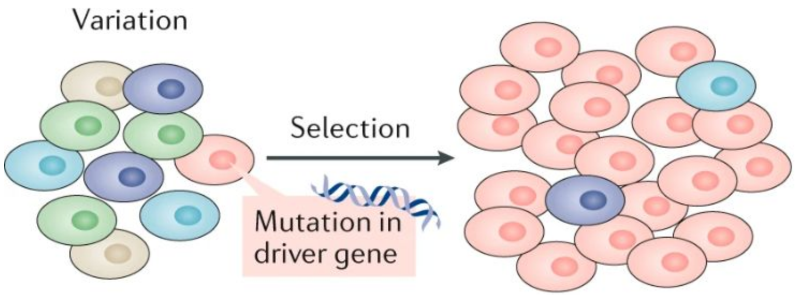
Somatic mutations and clonal selection

Ferriol Calvet
PhD Student @ IRB Barcelona (Nuria Lopez-Bigas' lab)

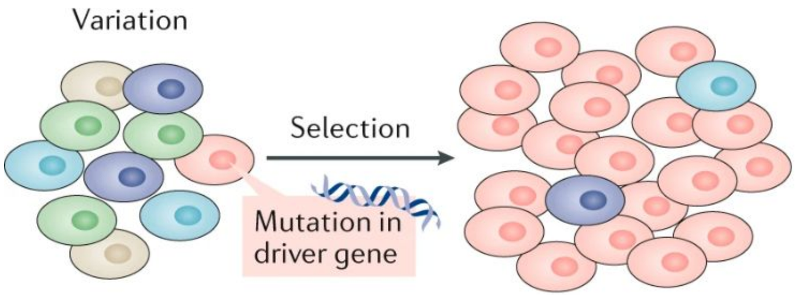
Cancer as an evolutionary process

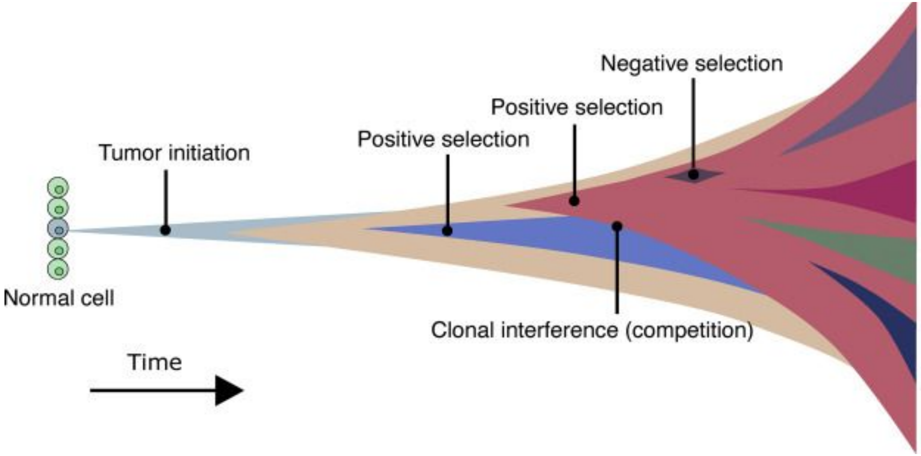


Darwinian evolution principles…

Figure from: Martinez-Jimenez et al. *Nat. Rev. Cancer* (2020)

Cancer as an evolutionary process

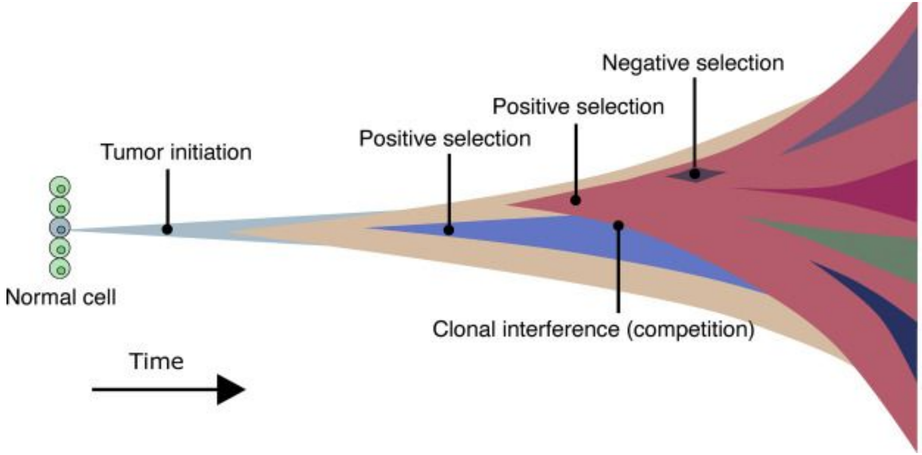




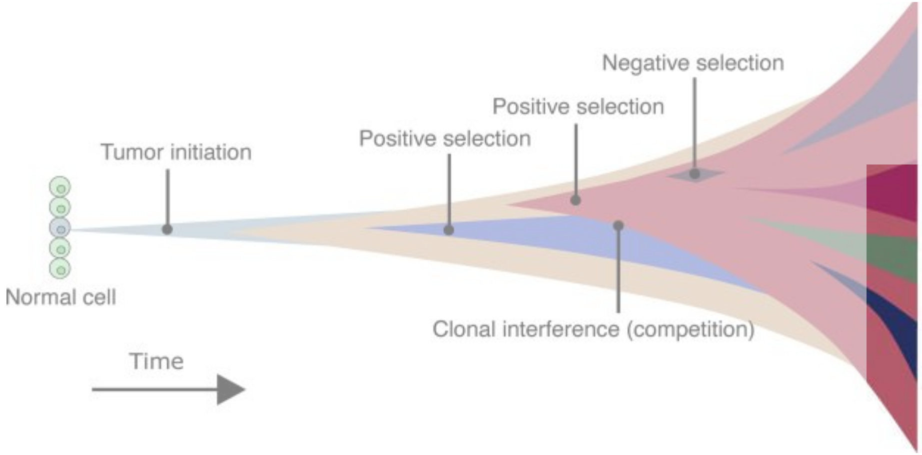
Darwinian evolution principles… … applied to tumorigenesis

Figure from: Lu et al. *Journal Genetics & Genomics* (2021)

Tumors are a final product of evolution

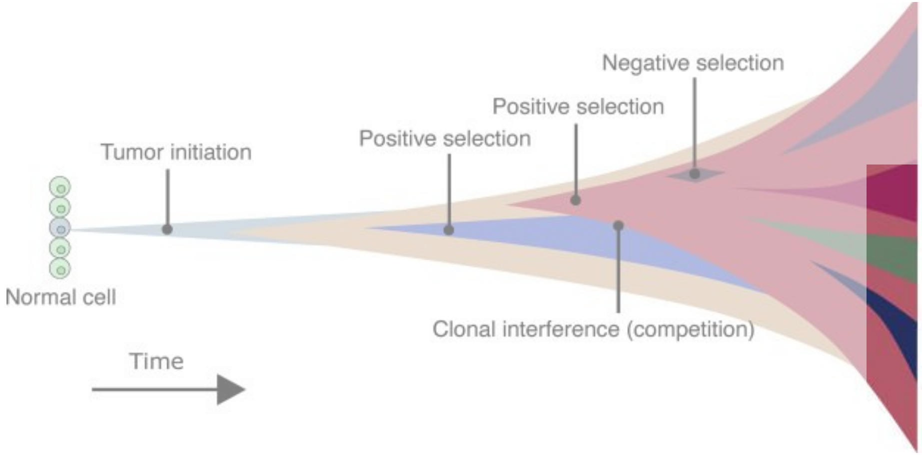


Tumors are a final product of evolution



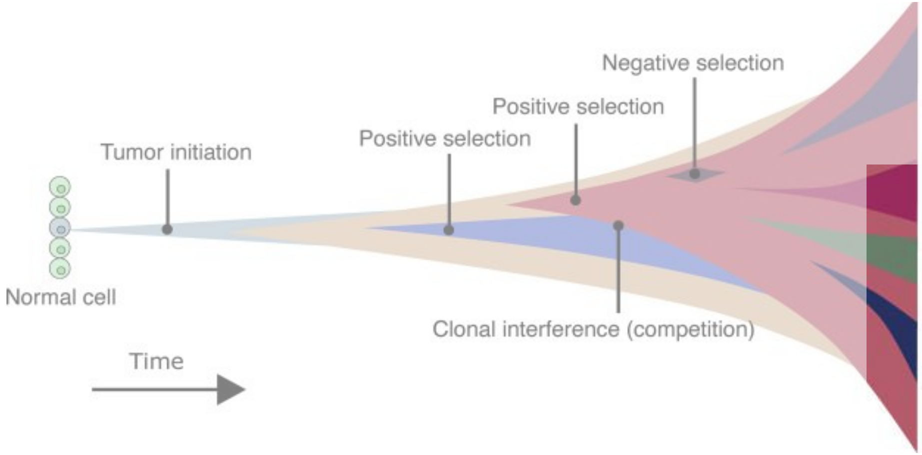
and also it is often the only piece of information we obtain…

We then extract as much information as possible…



- Identify mutational signatures
- Identify driver genes
- Timing signatures and driver events
- Estimate tumor heterogeneity
- …

Cancer epidemiology using somatic mutations



Identify mutational signatures

Identify driver genes

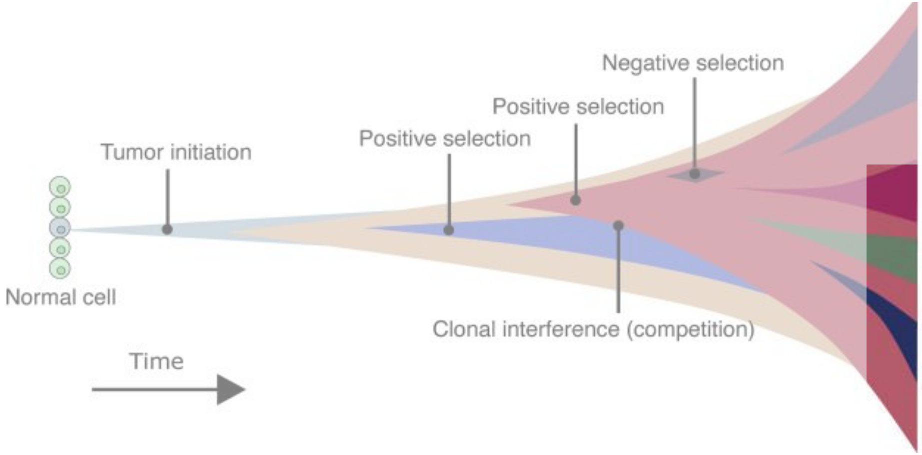
Timing signatures and driver events

Estimate tumor heterogeneity

…

These can inform tumor initiation and cancer risk,

Cancer epidemiology using somatic mutations



Identify mutational signatures

Identify driver genes

Timing signatures and driver events

Estimate tumor heterogeneity

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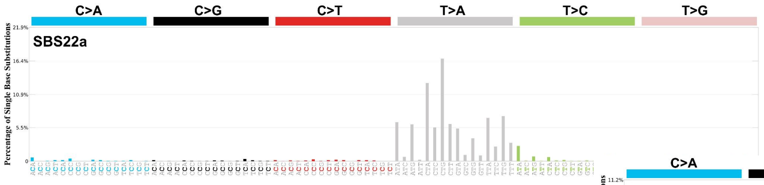
These can inform tumor initiation and cancer risk,

But to what extend?

Mutagenic carcinogens



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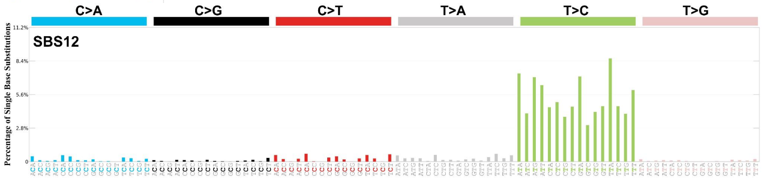
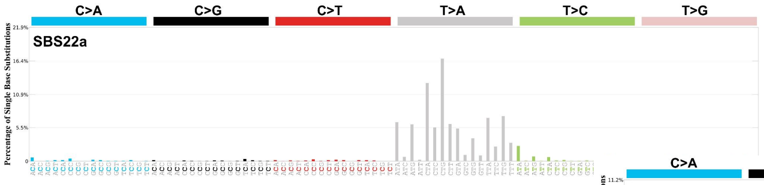


There are still non-mutagenic carcinogens…

Mutagenic carcinogens



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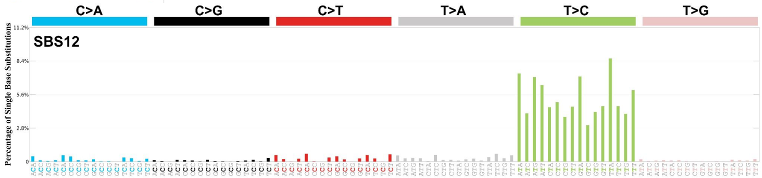
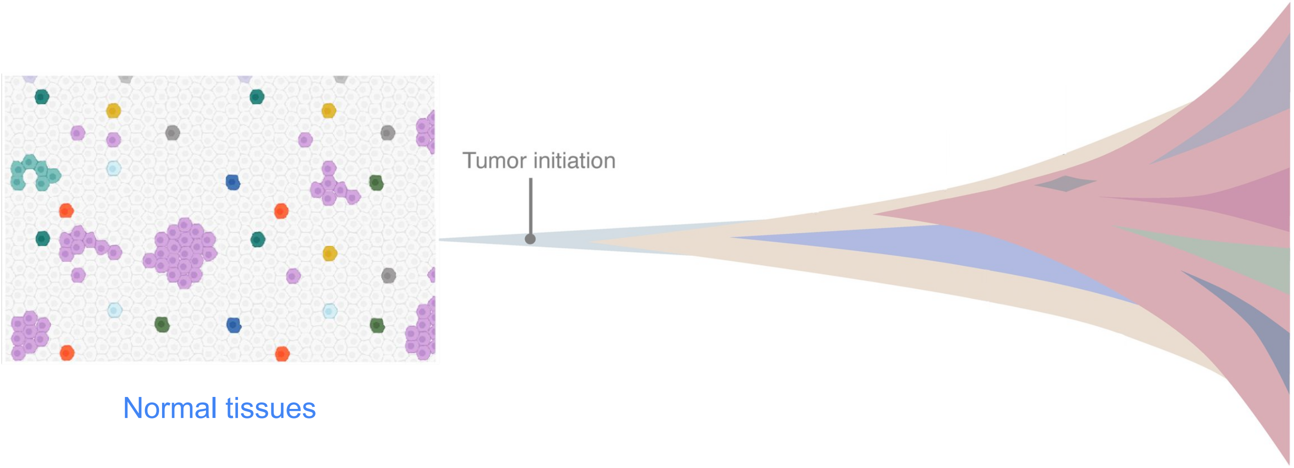


- World-wide variation in cancer incidence **cannot be explained by differences in the mutational processes**. *Moody et al. Nat. Genetics (2021)*
- Exposure to known human carcinogens leads to tumors in mice, but these **do not have more mutations** than spontaneous tumors. *Riva et al. Nat. Genetics (2020)*

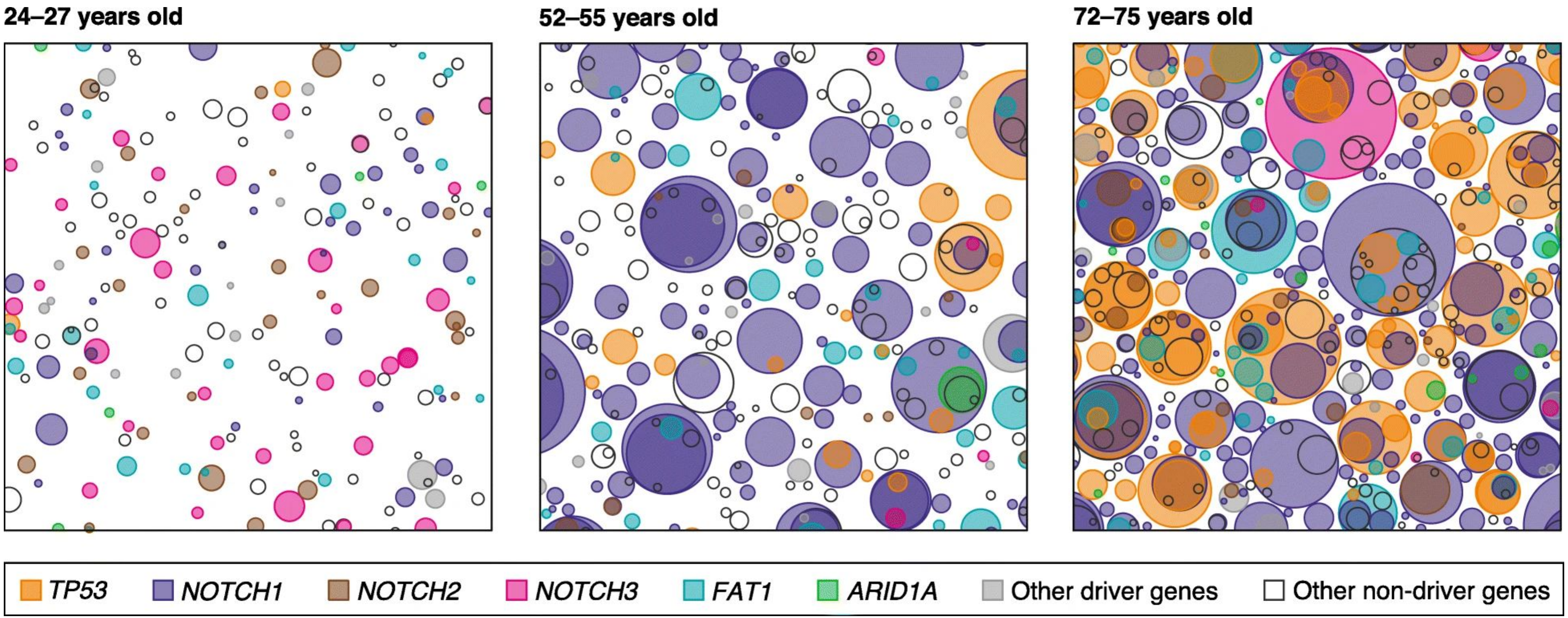
How can we detect the promotional carcinogens?



Normal tissues

& measuring clonal selection

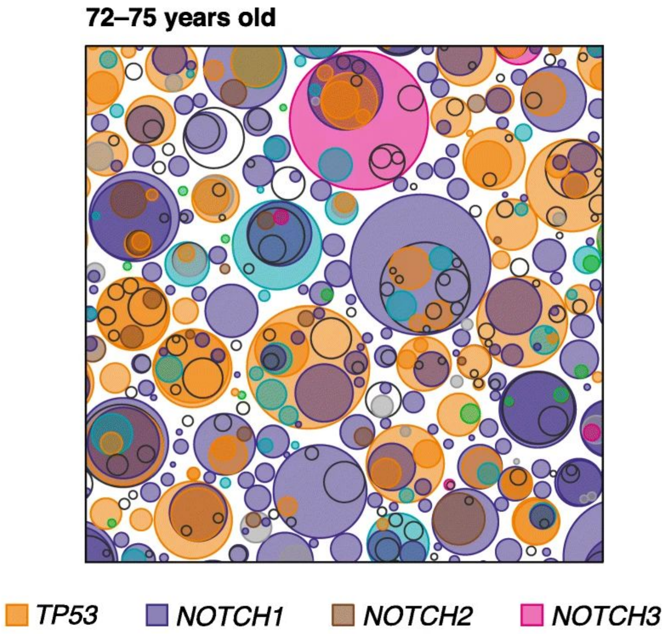
Cancer driver genes are mutated in healthy tissues



Human esophagus

Martincorena, I. *Genome Medicine* (2019)

Similar pattern to the one seen in tumors



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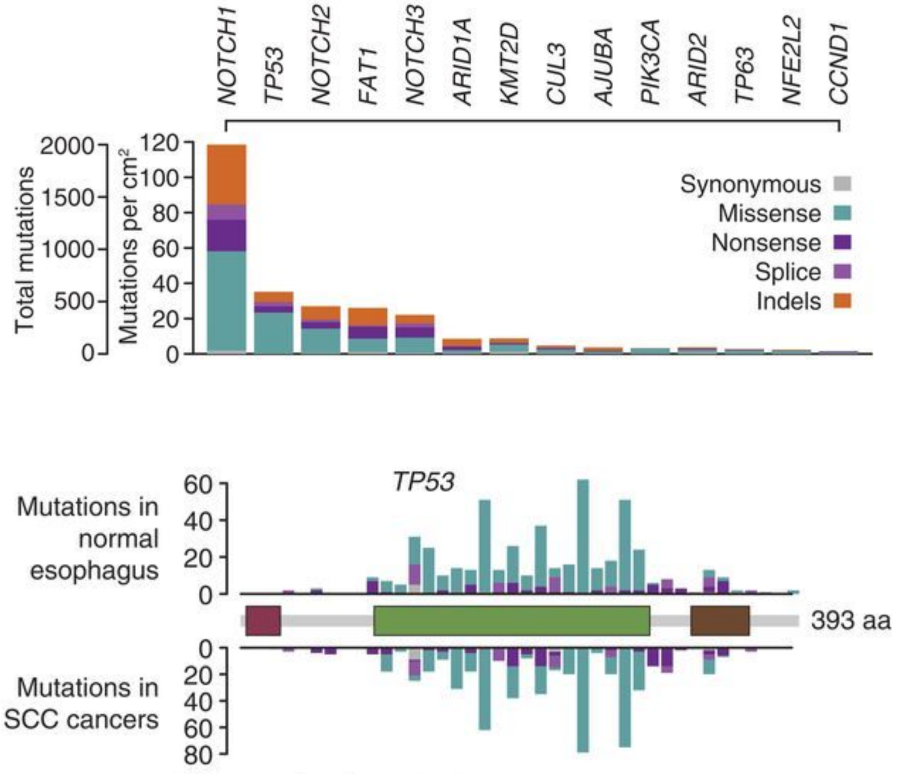
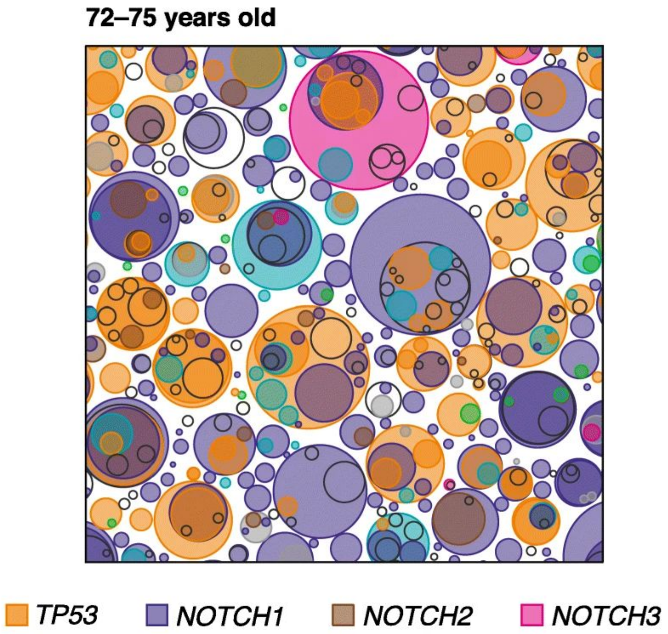


Martincorena et al. *Science* (2018)

Clear signals of selection for specific genes



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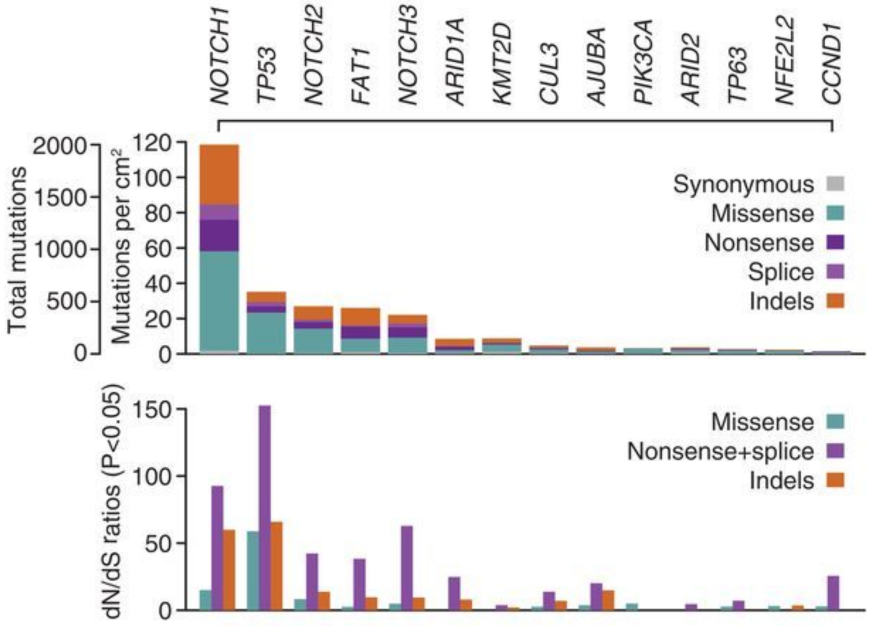
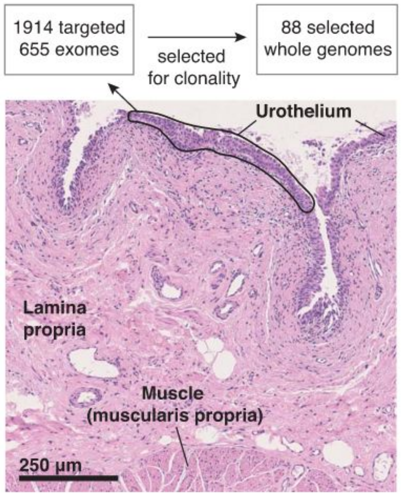


dN / dS ratios

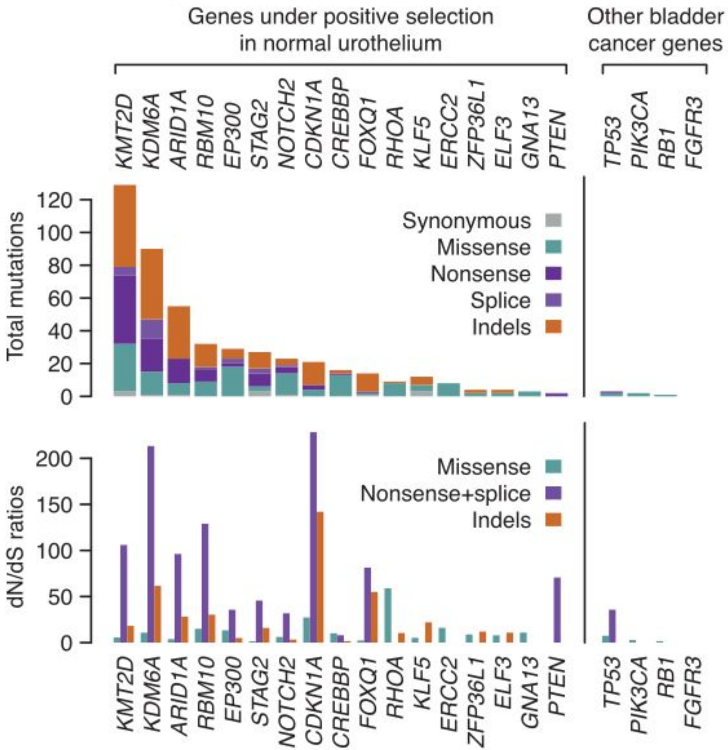
Martincorena et al. *Science* (2018)
Martincorena et al. *Cell* (2017) – dNdScv

Similar results were found across several tissues

Extensive heterogeneity in somatic mutation and selection in the human bladder



>1500 LCMs (15 donors)



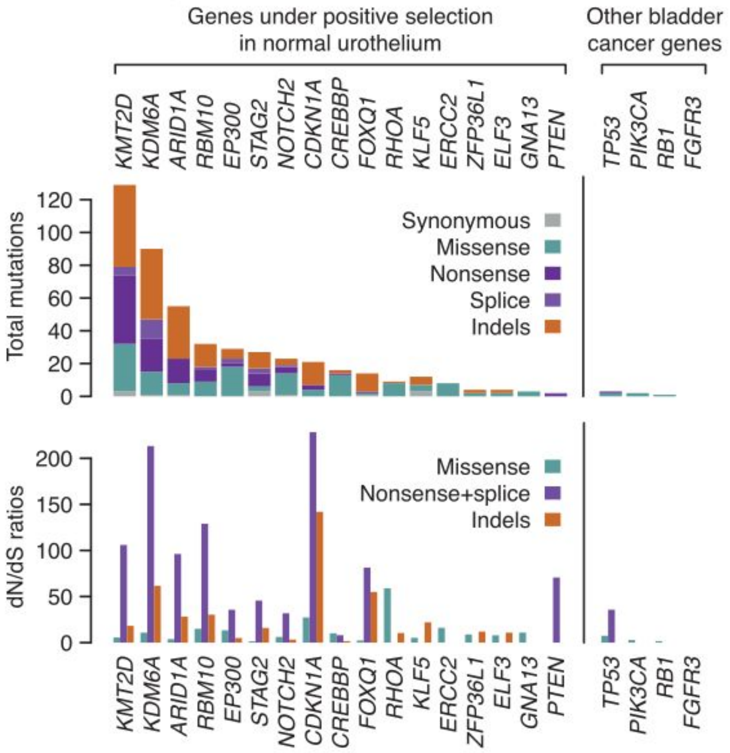
Lawson et al. *Science* (2020)

Differences between individuals, why?

Extensive heterogeneity in somatic mutation and selection in the human bladder

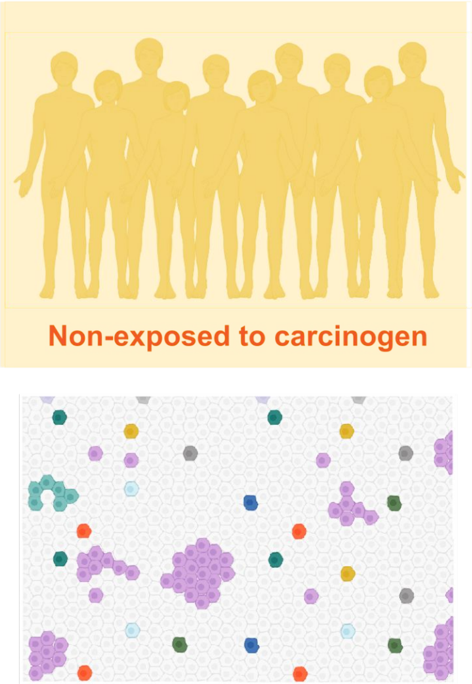
Donor	T01_25F	T02_35F	T03_53F	T08_61F	T10_68F	T11_69F	T13_71F	T14_75F	T15_78F	T04_55M	T05_58M	T06_59M	T07_59M	T09_61M	T12_70M
	Female									Male					
KMT2D	14	7	11	28	5	12	14	3	9	0	2	3	14	5	0
KDM6A	1	1	35	17	0	0	14	4	6	3	1	4	4	2	1
ARID1A	1	3	2	1	0	1	4	0	1	3	2	20	5	10	1
RBM10	0	0	5	1	1	1	1	1	0	1	1	1	13	6	0
EP300	2	1	6	5	1	2	3	0	1	0	0	2	1	4	1
STAG2	0	0	3	7	4	0	5	0	1	0	1	1	1	1	0
CDKN1A	0	4	2	6	0	0	0	0	1	3	0	5	0	1	0
NOTCH2	0	1	7	5	0	1	0	0	2	1	0	2	1	2	0
CREBBP	0	0	1	5	0	2	1	1	2	3	0	1	0	0	0
FOXQ1	0	0	5	1	1	0	1	0	0	0	1	0	2	0	2
Mutations	65	49	196	161	34	73	105	31	66	42	17	101	81	70	60
Samples	105	93	88	200	31	122	102	36	59	38	17	76	76	72	73

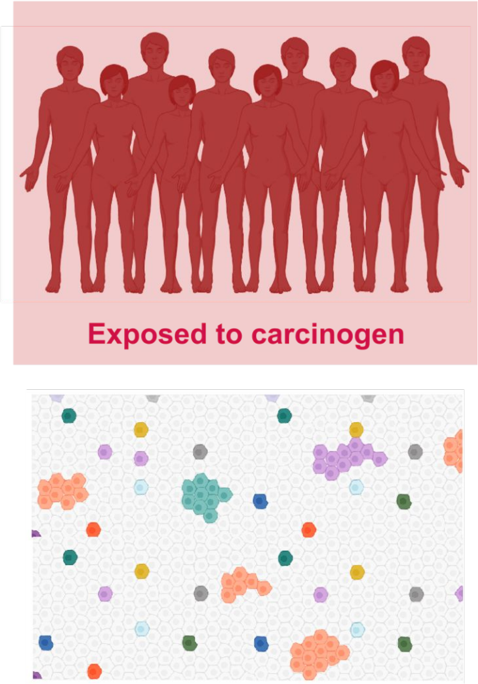
Number of unique mutations 0 5 10 15 20 25 30 35



Lawson et al. *Science* (2020)

Applying CGE in normal tissues…





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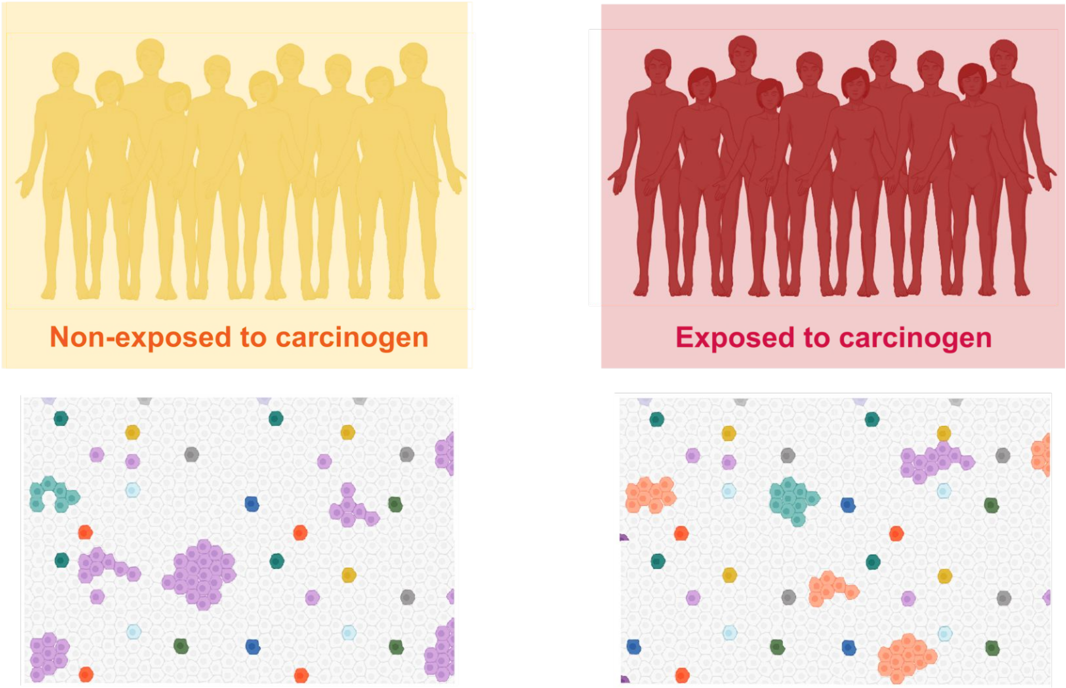

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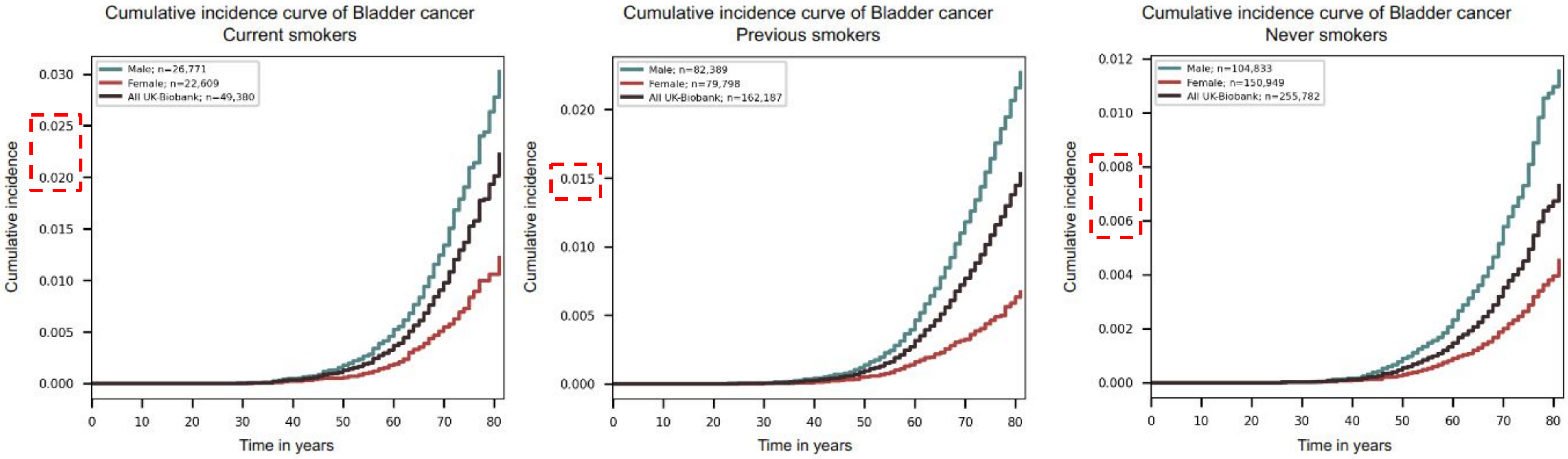
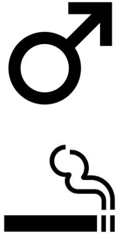
- Risk factors
- Cohort
- Measurements

Bladder cancer risk factors: sex and smoking



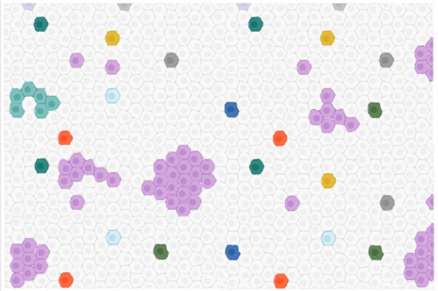
Males and smokers have an independent higher risk of bladder cancer

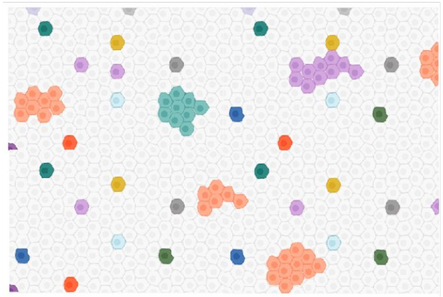
Can we see differences between the groups?



Males and smokers have a higher risk of bladder cancer

Clonal landscape





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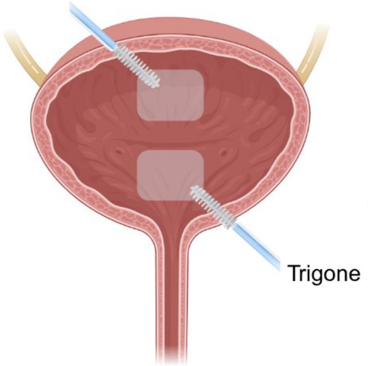
Deep profiling of the normal bladder clonal structure



1. Broad sampling

Cytobrushing (2 cm²)

(79 samples)

UW Medicine

45 non-bladder cancer donors (autopsies)

Calvet, Blanco *et al*. (Nature, 2025)

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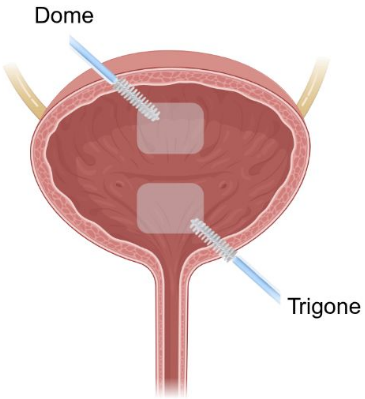

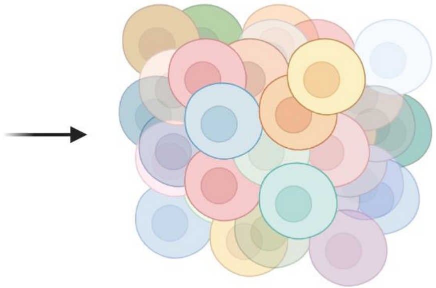

Mixture of clones
(= better representation of the tissue)

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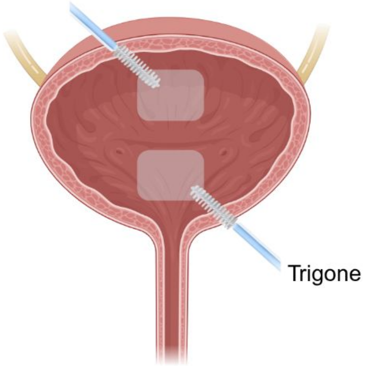

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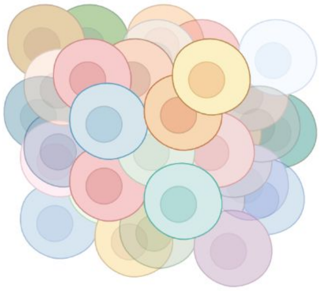

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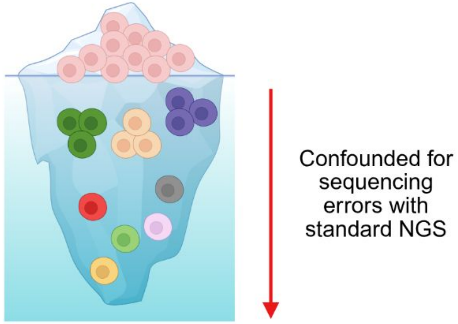


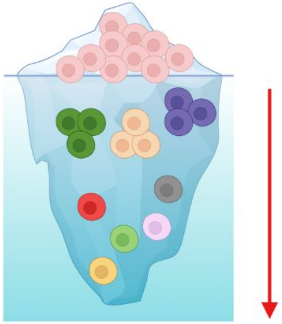
2. Deep sequencing

(= detecting mutations in very small clones)

Error-corrected duplex-sequencing

Calvet, Blanco *et al*. (Nature, 2025)

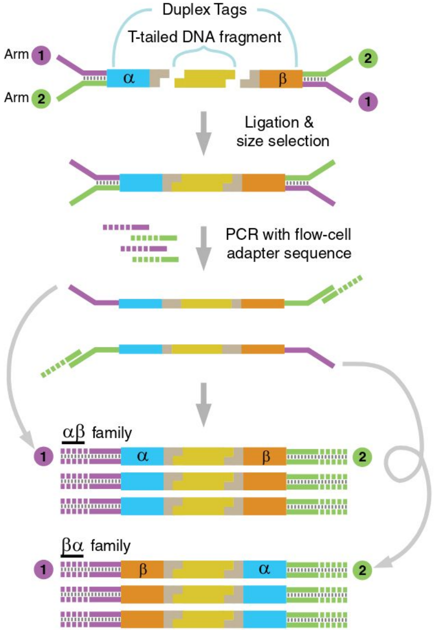
Duplex sequencing enables deep sequencing

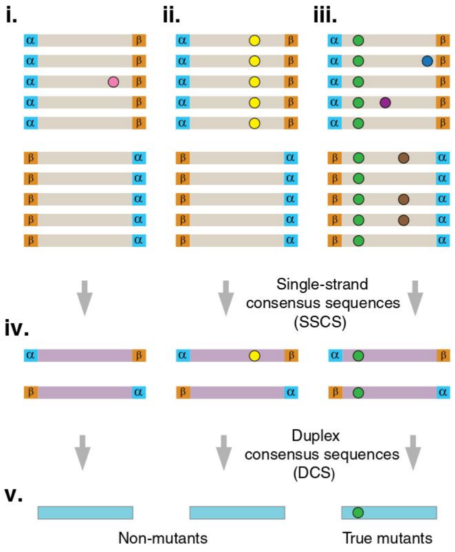


Confounded for sequencing errors with standard NGS

2. Deep sequencing

(= detecting mutations in very small clones)





Adapted from Schmitt et al. *PNAS* (2012)

Kennedy et al. *Nature Protocols* (2014)
Abascal et al. *Nature* (2021)
Lawson et al. *Nature* (2025)

…
Nandi et al. *bioRxiv* (2025)
Calvet et al. *protocols.io* (2026)

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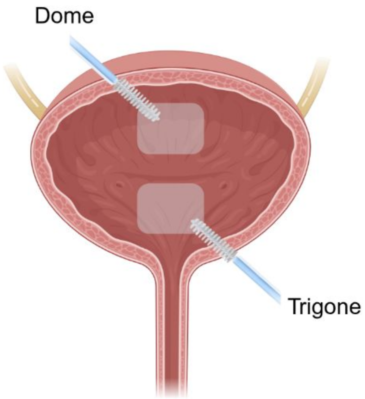

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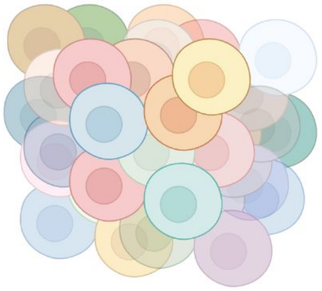
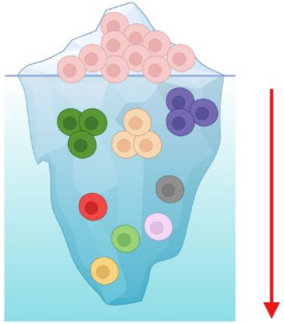


Mixture of clones (= better representation of the tissue)



Confounded for sequencing errors with standard NGS

2. Deep sequencing

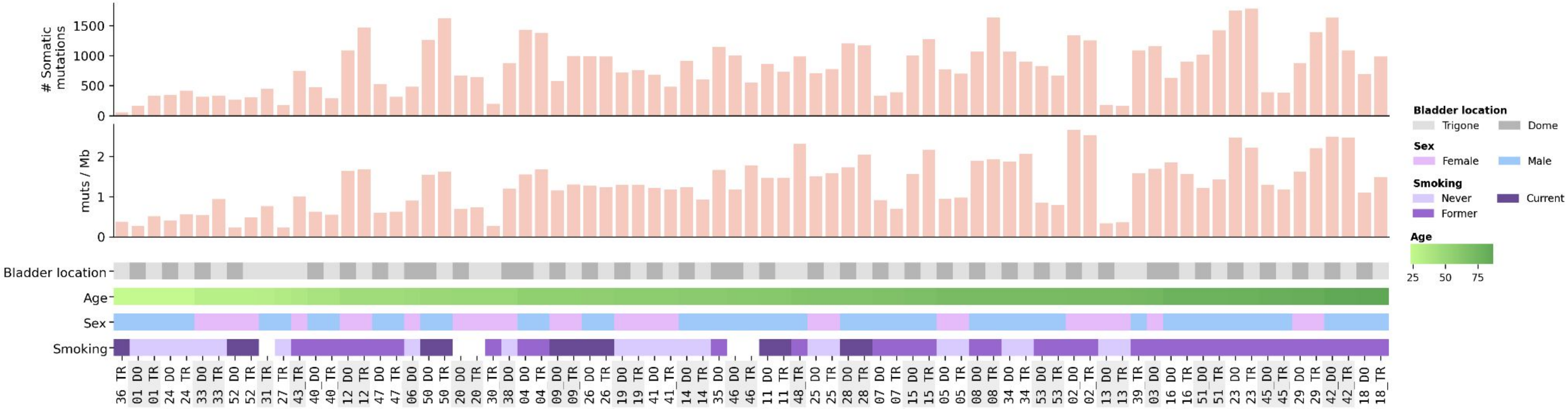
(= detecting mutations in very small clones)

Error-corrected duplex-sequencing

16 genes panel

Depth 5000x

Hundreds of mutations per sample



64,278 mutations in total, only with 79 samples.

How to compute clonal selection?

Compare observed mutations to those expected to occur neutrally.

dN / dS

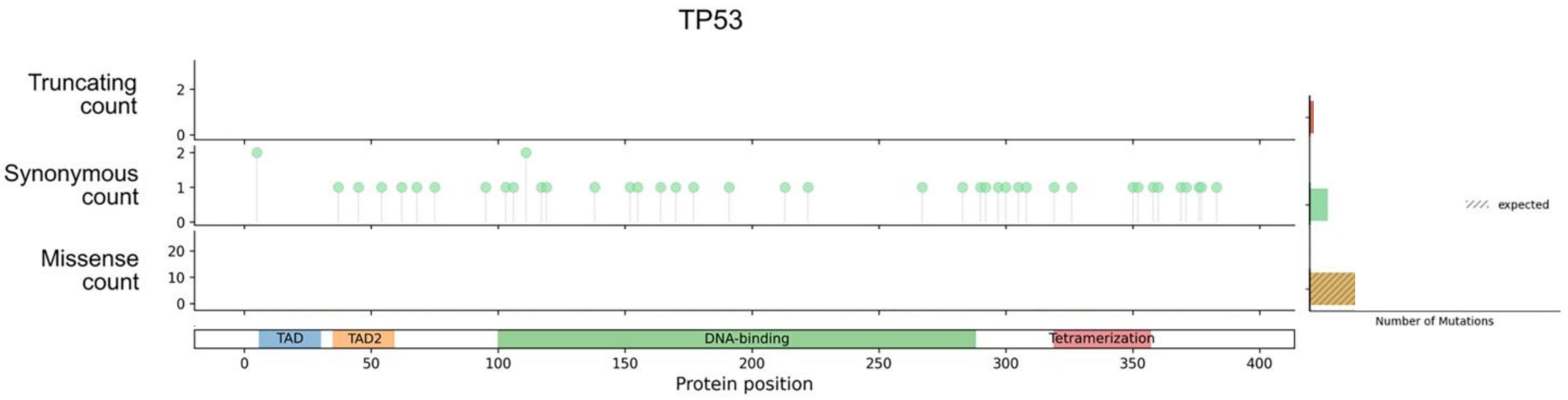
Functional impact bias

3D clustering

Computing dN/dS – excess of mutations

Estimate background mutation rate, apply it to the non-synonymous mutations.

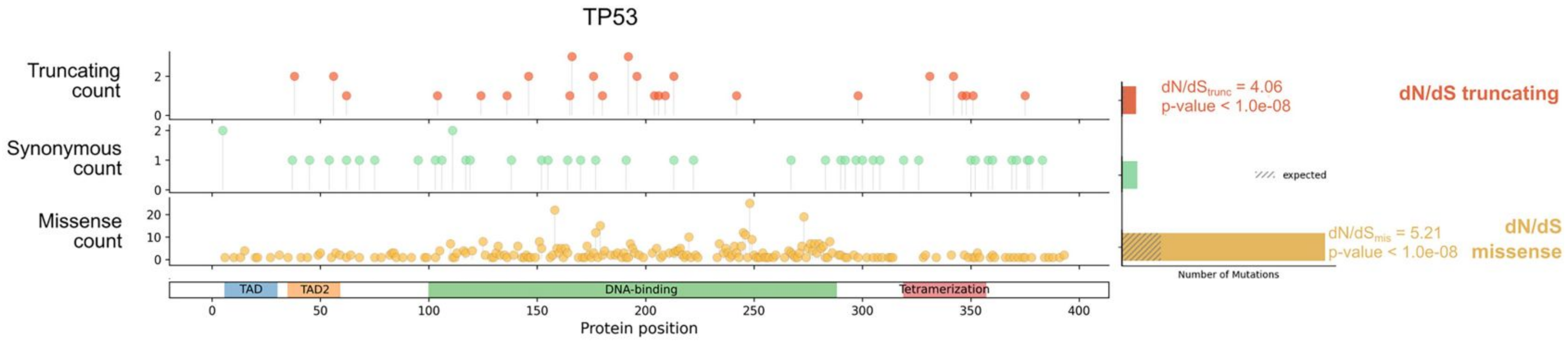
dN / **dS**



Computing dN/dS – excess of mutations

Compare the expected number of mutations to the observed ones.

dN / dS



Calvet, Blanco et al. *Nature* (2025) – omega
Based on Martincorena et al. Cell (2017) – dNdScv

Computing a functional impact bias

Measure, numerically, the consequence of all mutations in the genome.

Adapted version of Mularoni et al. *Genome Biology* (2016)

Computing a functional impact bias

Measure, numerically, the consequence of all mutations in the genome.

Scores table

#Chrom	Pos	Ref	Alt	Score
1	10001	T	A	7.993
1	10001	T	C	8.285
1	10001	T	G	8.08
1	10002	A	C	8.043
1	10002	A	G	8.244
1	10002	A	T	7.942
1	10003	A	C	8.16
1	10003	A	G	8.252
1	10003	A	T	7.963
1	10004	C	A	6.694
1	10004	C	G	6.836
1	10004	C	T	7.164
1	10005	C	A	6.731
1	10005	C	G	6.834
1	10005	C	T	7.168

Adapted version of Mularoni et al. *Genome Biology* (2016)

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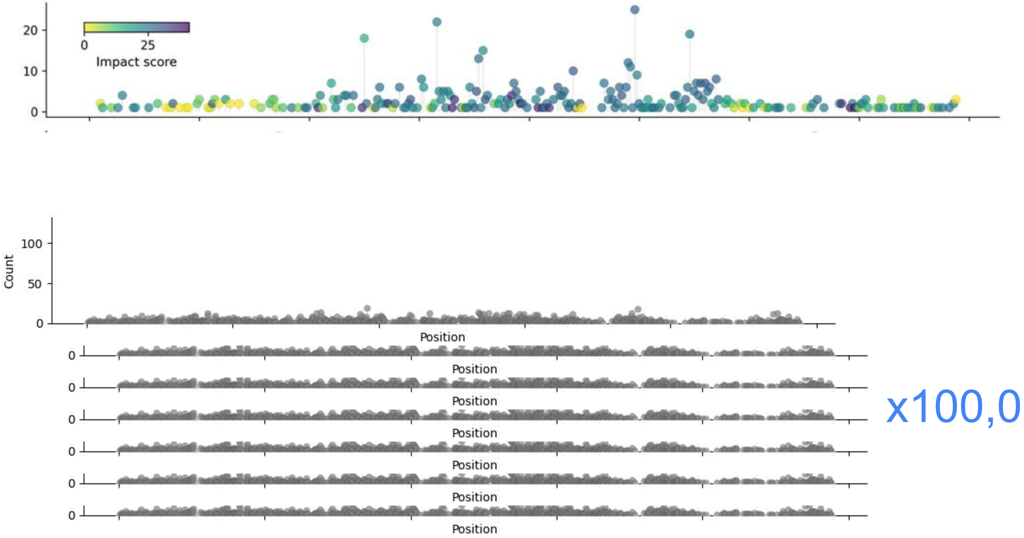
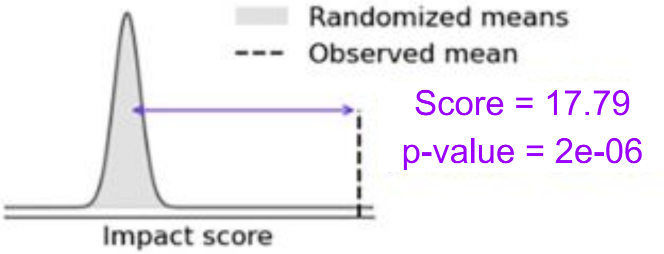


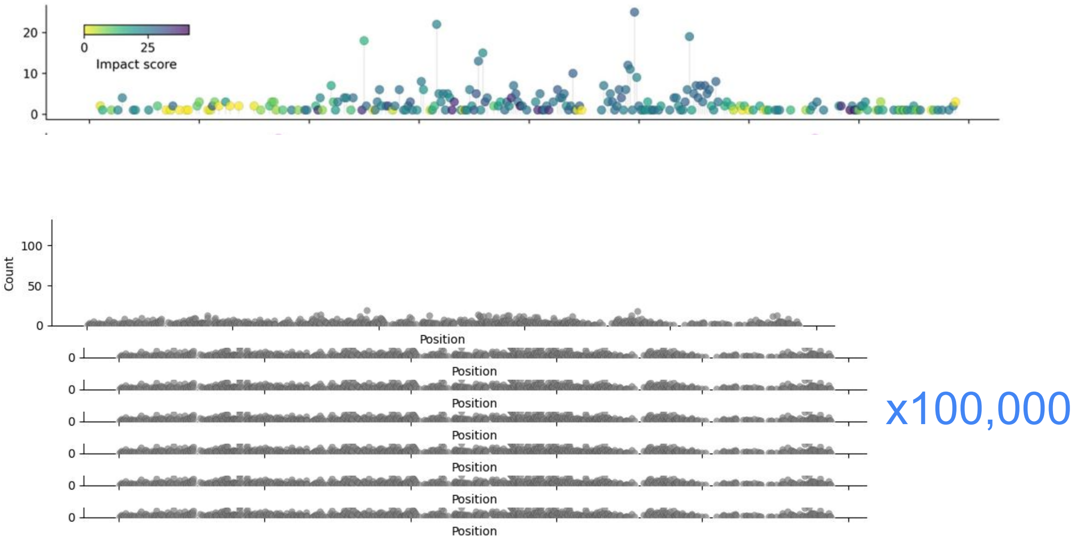
x100,000

Adapted version of Mularoni et al. *Genome Biology* (2016)

Computing a functional impact bias

Compare the observed value with the random distributions of mutations.

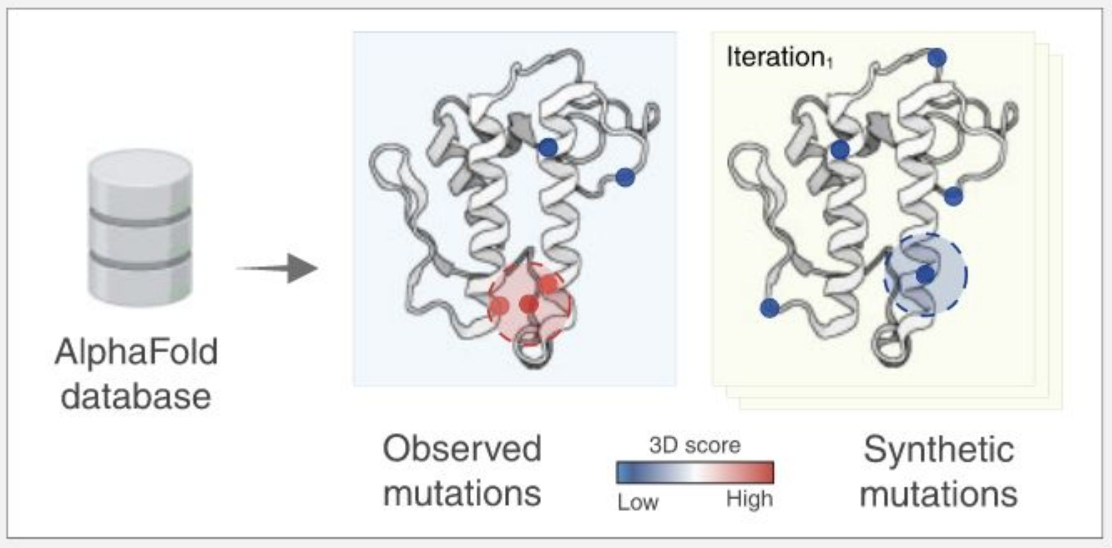




Adapted version of Mularoni et al. *Genome Biology* (2016)

Identify clusters of mutations in the 3D space

Leveraging AlphaFold and the expected distribution of mutations…



Pellegrini et al. *Nucleic Acids Research* (2025)

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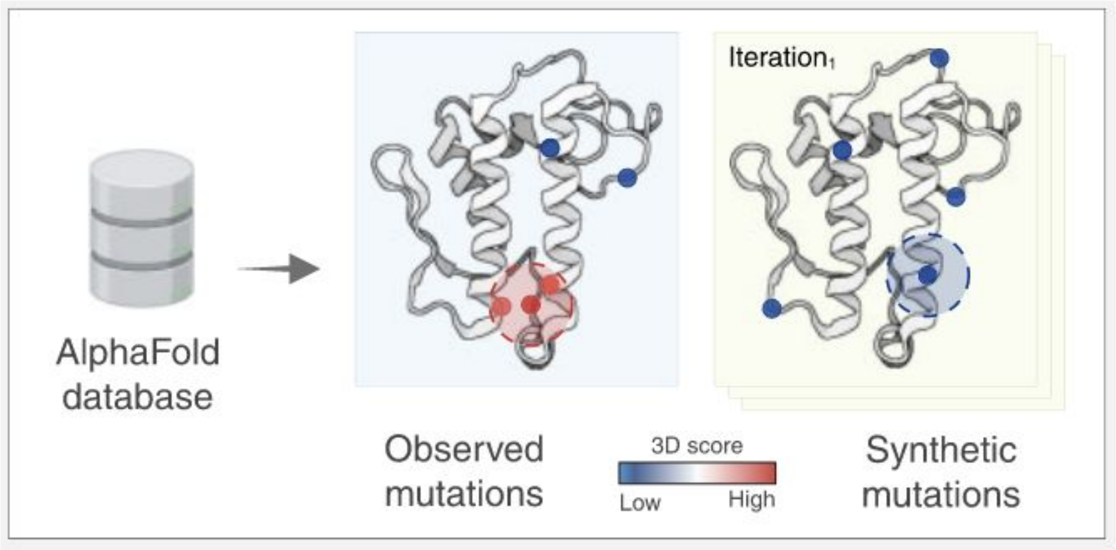
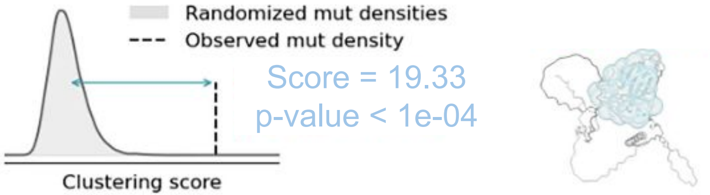


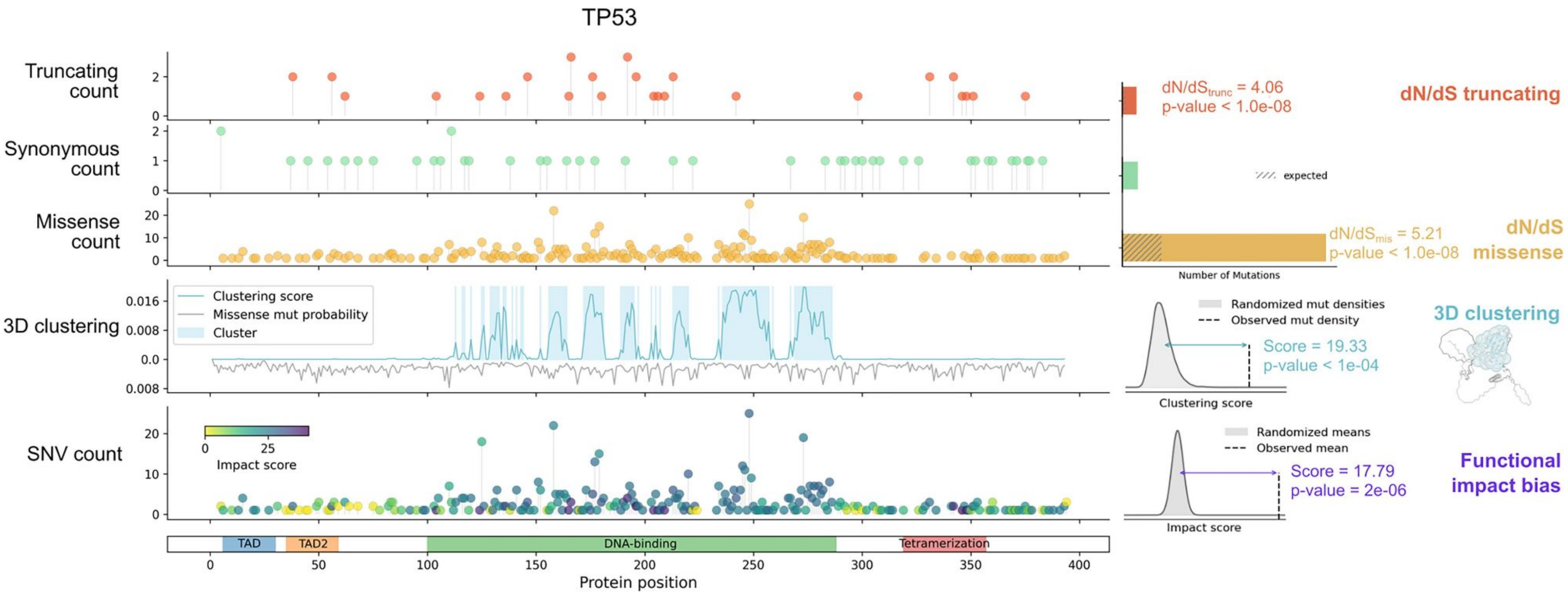
3D clustering



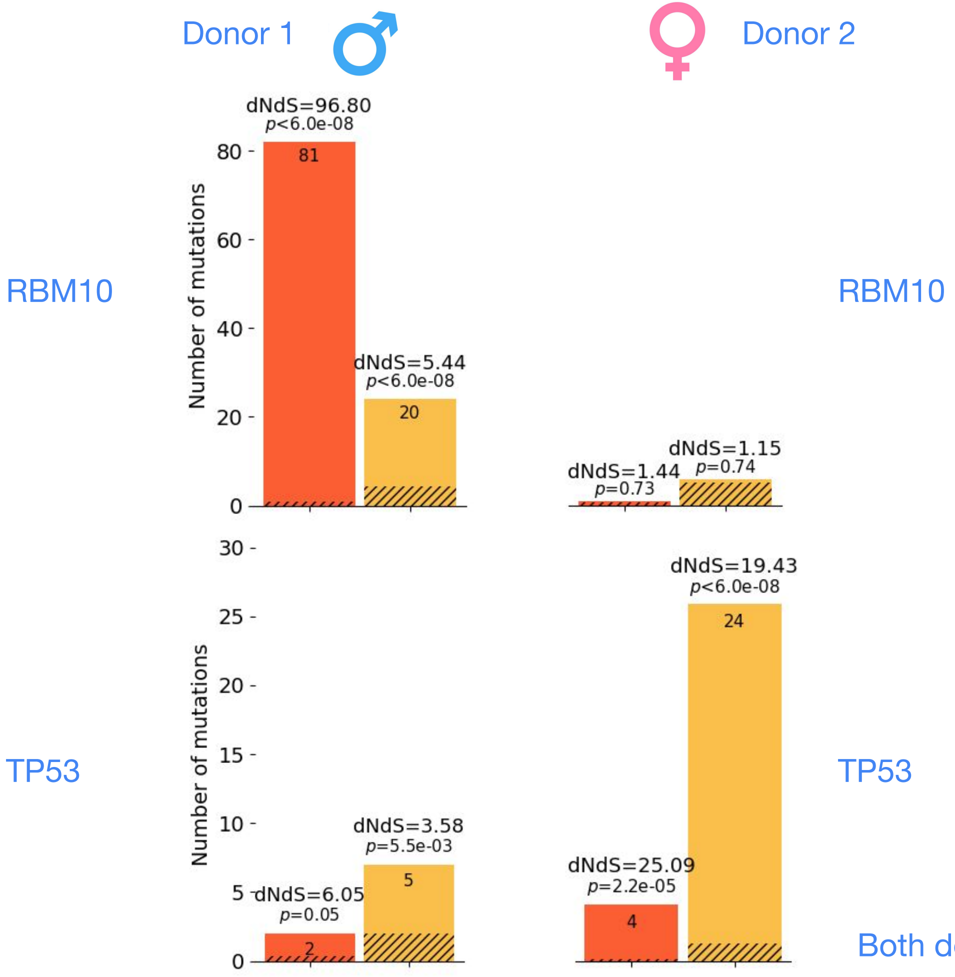
Pellegrini et al. *Nucleic Acids Research* (2025)

Multiple ways of computing clonal selection

dN / dS Functional impact bias 3D clustering



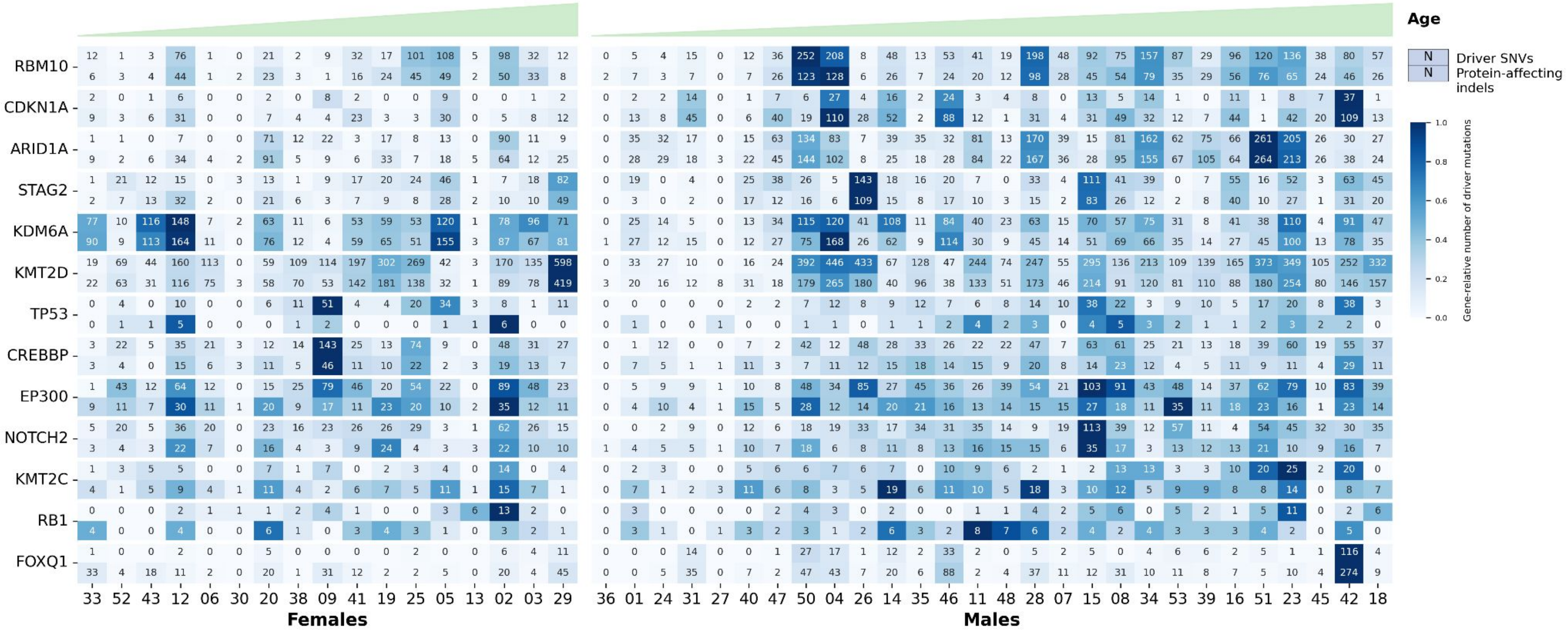
Enough power to compute positive selection at the sample level



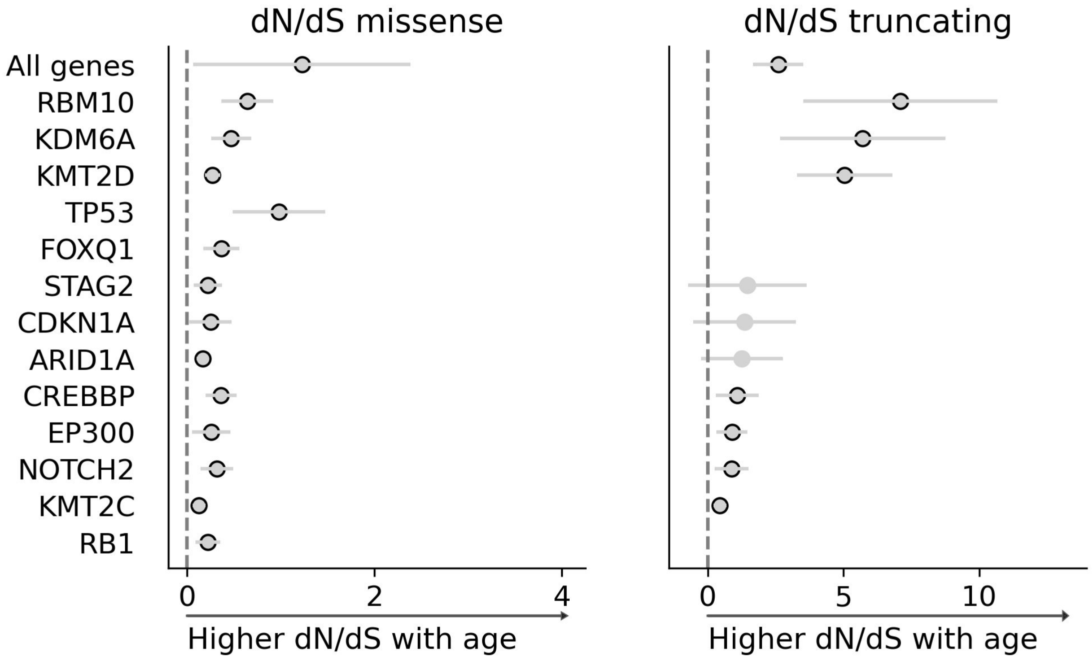
Focussing on **dN/dS** metric

Both donors were 56 years old

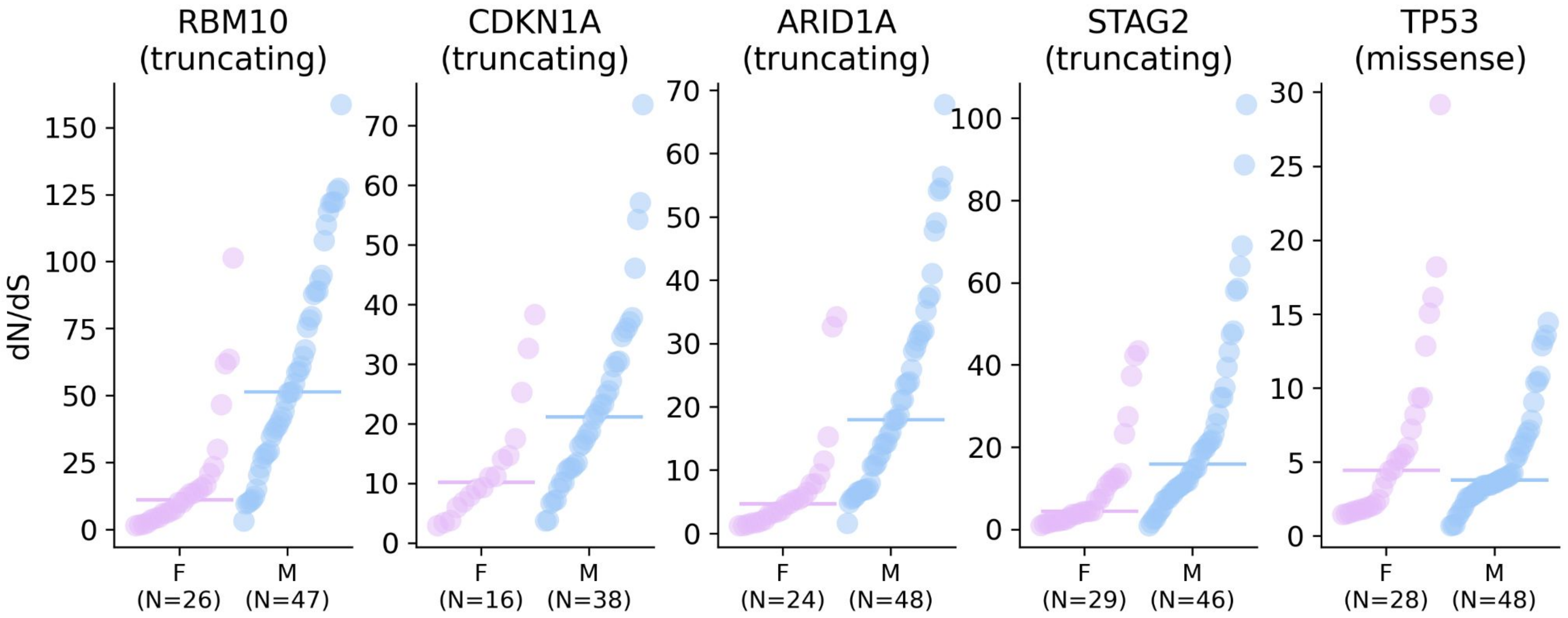
Interindividual variability in the magnitude of positive selection



Increased positive selection with age



Increased positive selection for males in some genes



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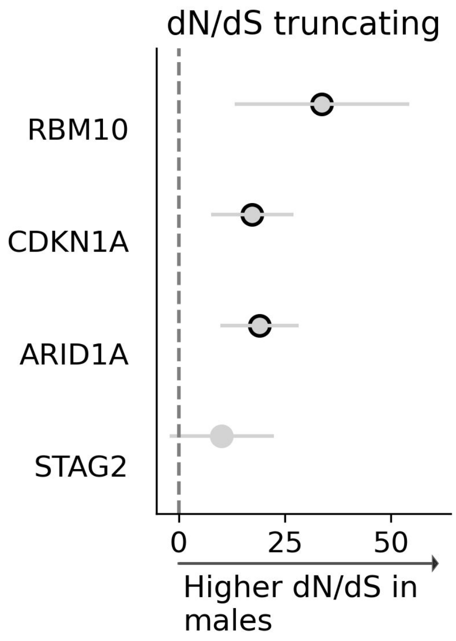
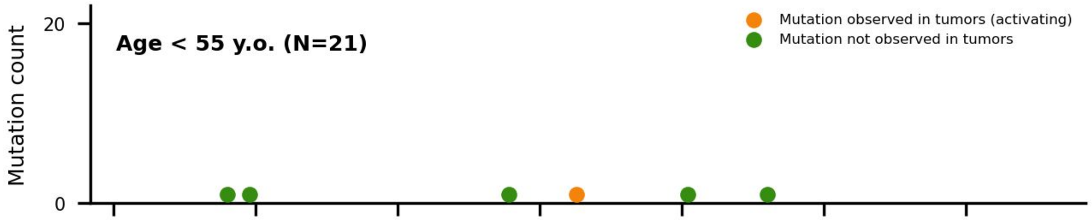


Mutations in the TERT promoter

TERT is responsible for elongating the telomeres, constitutive activation can make the cells immortal.

Mutations in the TERT promoter



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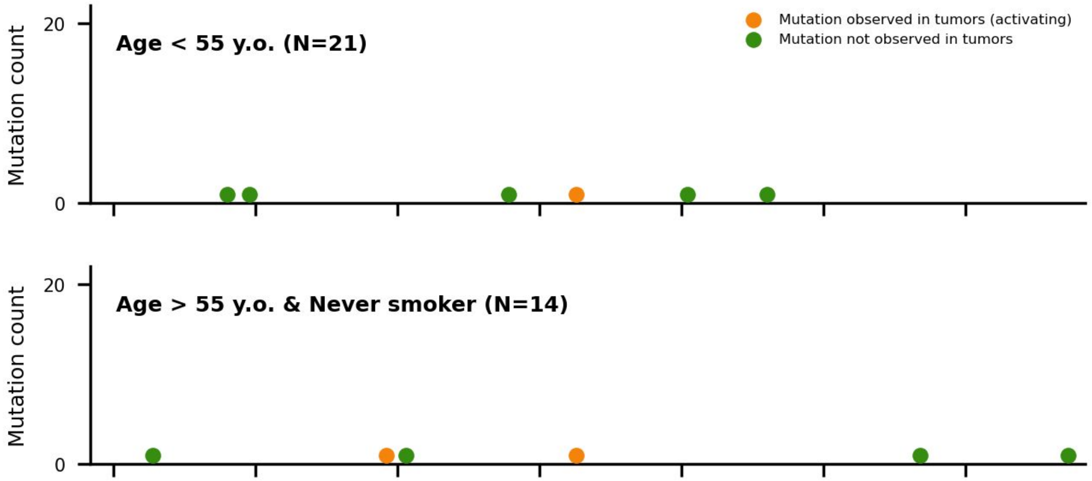
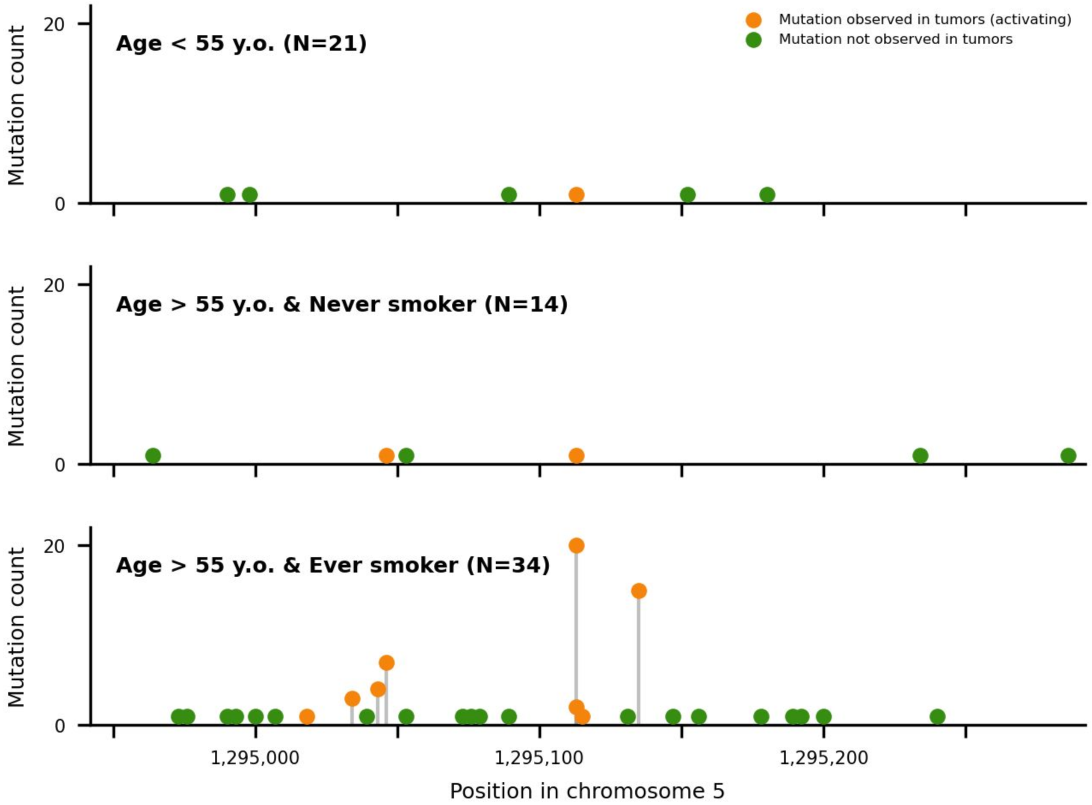
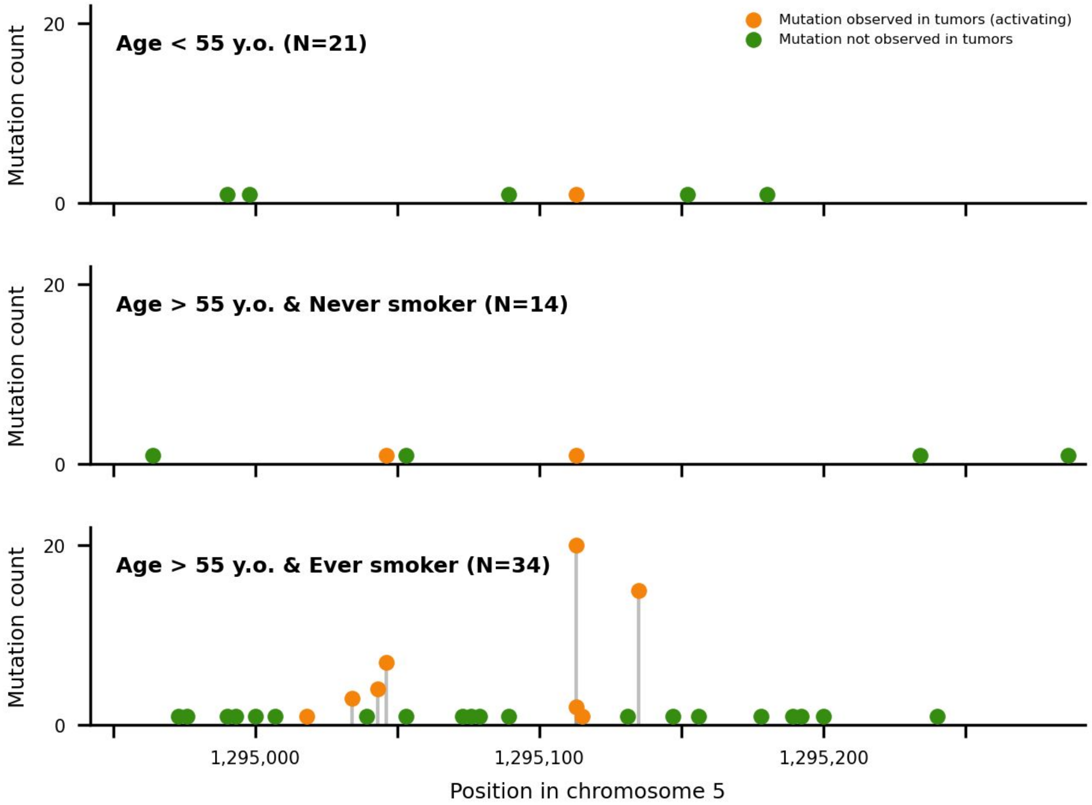
Mutations in the TERT promoter

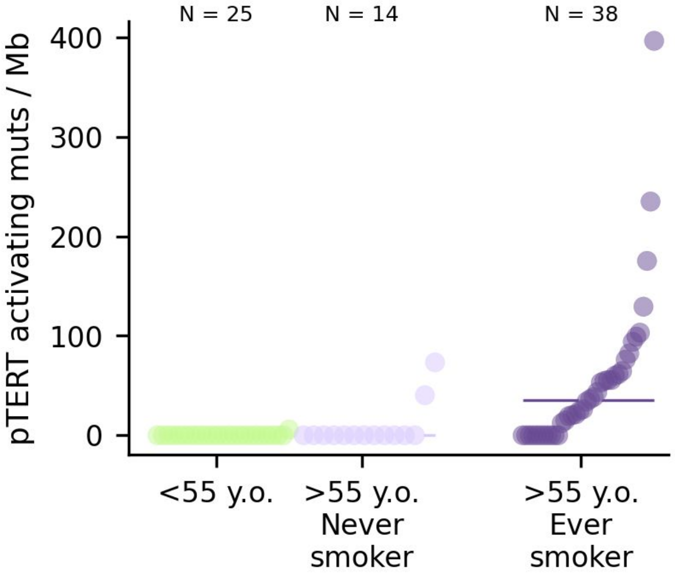


Mutations in the TERT promoter

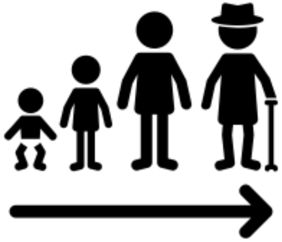


Increased TERT selection in smokers >55 y.o.



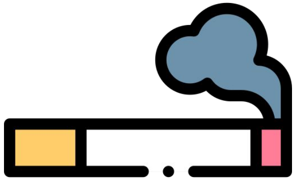


How does this relate to cancer risk?



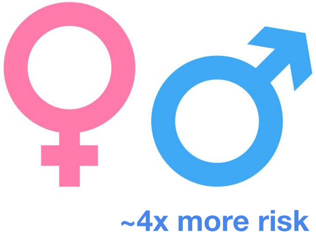
Aging

Age increases the number of mutations and clone sizes



Smoking

Increases the prevalence of TERT promoter mutations. Very frequent in bladder cancer.

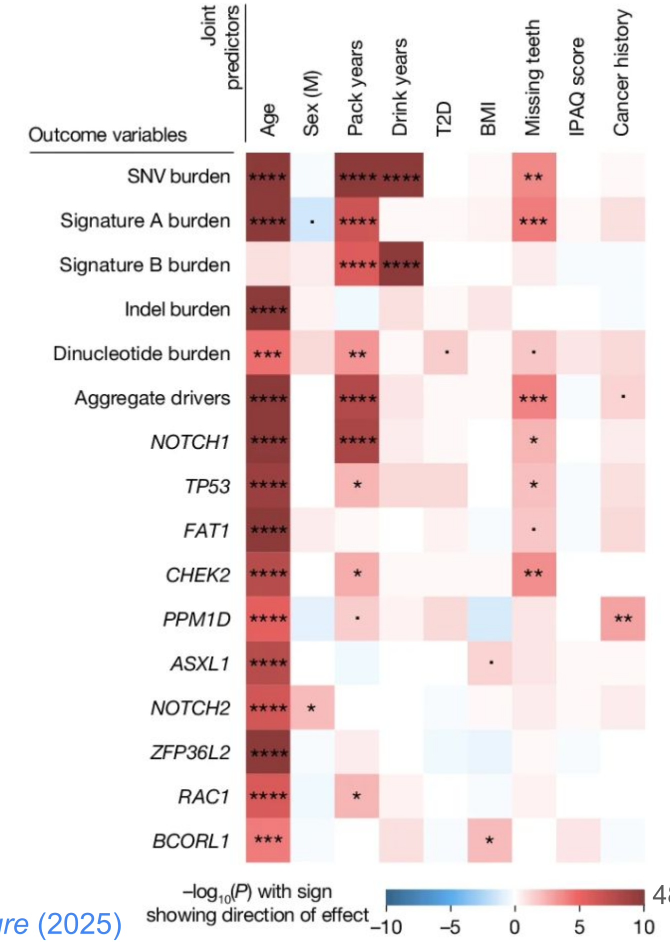


Sex

Males have higher selection of 3 genes not amongst the most frequent bladder cancer drivers.

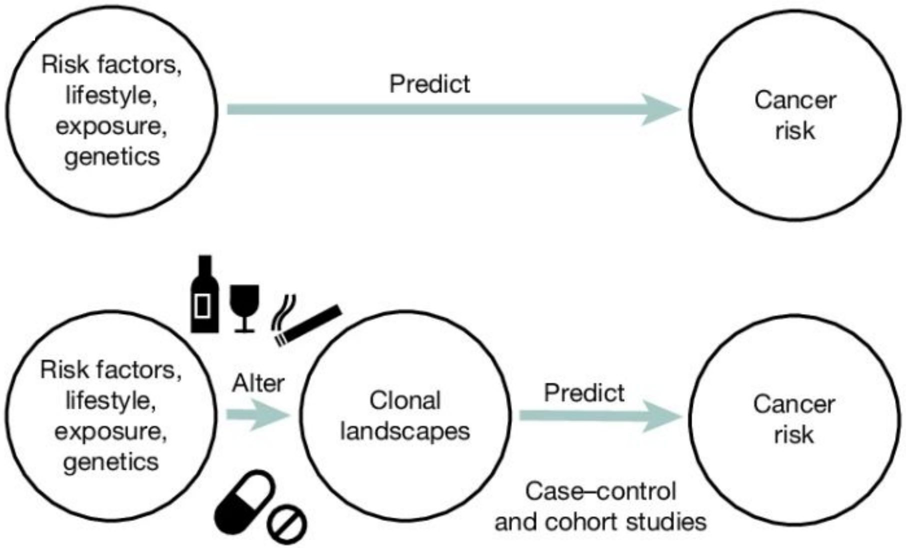
Other examples of exposures

- Buccal swabs
  - Smoking favours the selection of NOTCH1 and TP53. (see figure)
- Air pollution and lung cancer.
  - PM2.5 levels and EGFR expansions.
- Chemotherapy & blood cancer and CH
  - Clones with mutations in DNA damage repair genes expand with chemotherapy.



From Lawson et al. *Nature* (2025)

Clonal landscapes between exposures and cancer risk



From Lawson et al. *Nature* (2025)

Take home messages

- Normal tissues inform about the evolution prior or leading to tumor initiation.
- Clonal selection measures the result of mutations followed by certain selective pressures, and can be quantified in multiple ways.
- Exposure to some carcinogens or cancer risk factors can modify the selective pressures on this normal tissues, which may or may not increase cancer risk.

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Thank you for your attention!
Questions?

and thanks to …

IRB BARCELONA

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BARCELONA BIOMEDICAL GENOMICS LAB

CANCER GRAND CHALLENGES
Team: PROMINENT